

BrickAtlas: Auditing Display-Grounded Evidence Following with Paired Observations of LDraw Parts

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Abstract

001 Answering a visual question correctly does not establish
002 that a model tracks the evidence selected by its reference.
003 A paired test can address this gap only if the reference and
004 the answer remain well defined after the intervention. We
005 introduce a display-grounded observation contract on 24
006 human-authored LDraw assemblies. Numeric labels select
007 objects in the current image; canonical source identities
008 provide audit provenance, not immutable answers to edited
009 images. Two separate families test rendered color changes
010 at fixed geometry (73 pairs) and candidate-reference reas-
011 signment for geometry matching (67 pairs). A finite assign-
012 ment balances choice transitions and bounds 13 declared
013 oracle-A rules before inference. These privileged audits sup-
014 ply A gold; they neither describe legal single-observation
015 attacks nor bound models. Human assignment prevents any
016 reviewer seeing two views of one parent. A frozen three-
017 model roster binds every planned run and failure to analysis.
018 A supporting graph audit demonstrates why the choice of
019 control matters: an isolates-plus-one heuristic solves 72/73
020 natural questions and all 61 degree-visible change pairs,
021 but has zero both-correct on 12 degree-matched pairs; BFS
022 solves all of them. The strong subset is a small proof of con-
023 cept, including an anonymous 1-WL blind spot. We release
024 machine-verified stimuli, the executable evaluation proto-
025 col, and a searchable 617-item historical dossier. Human
026 visual certification and learned-model diagnostic value re-
027 main empirical requirements; the reported findings concern
028 stimulus and algorithmic audits.

029 1. Introduction

030 A label may identify a source instance or select a visible ob-
031 ject. These meanings diverge under intervention: relabeling
032 another part B0035 does not change the original B0035's
033 type. Changing gold without specifying the referent risks
034 rewarding interpretation shifts.

035 We ask a narrower, observable question: *does a response*

correctly follow the property or candidate reference se- 036
lected in the current display, when that observation changes 037
under an explicit contract? The contract resolves numeric 038
references within the supplied image. The source identity 039
is retained for construction, pairing and inspection. The 040
model is never asked to recover a historical source fact from 041
a deliberately edited display. 042

We study two visual families separately (Figure 1). 043
Color changes rendered appearance at fixed geometry, pose 044
and camera. Part-type fixes the target while two candidate 045
labels exchange referents: the correct token changes with- 046
out changing a canonical part's type. The families support 047
different interpretations and no pooled visual score. 048

Paired image-dependent answers have substantial prece- 049
dent in VQA [3], contrast sets [2], and joint ground- 050
ing tests [12]. VisualFLIP directly studies independently 051
queried answer-flipping image pairs [14]. Pair accuracy, 052
Both and minimal image edits are not our contributions. 053
Our measurement contract links CAD provenance, current 054
display referents and exact-observation qualification: a cor- 055
rect source record alone does not establish a uniquely visi- 056
ble answer. 057

The separate graph audit shows how control construc- 058
tion changes interpretation: isolates-plus-one solves 72/73 059
natural questions and 61/61 degree-visible pairs, yet has 060
Both=0/12 on degree-matched pairs. BFS solves all of them. 061
This distinguishes the test distributions without identifying 062
a learned model's internal method. 063

We contribute (1) a display-grounded referent contract 064
separating source truth from current visual answers; (2) 065
pre-inference commitment to a restricted shortcut audit and 066
joint choice assignment; and (3) response transition analysis 067
derived from raw receipts and parent-disjoint human review. 068
The commitment is locally hash-locked, not externally reg- 069
istered. Whether these elements yield useful learned-model 070
diagnostics is empirical. 071

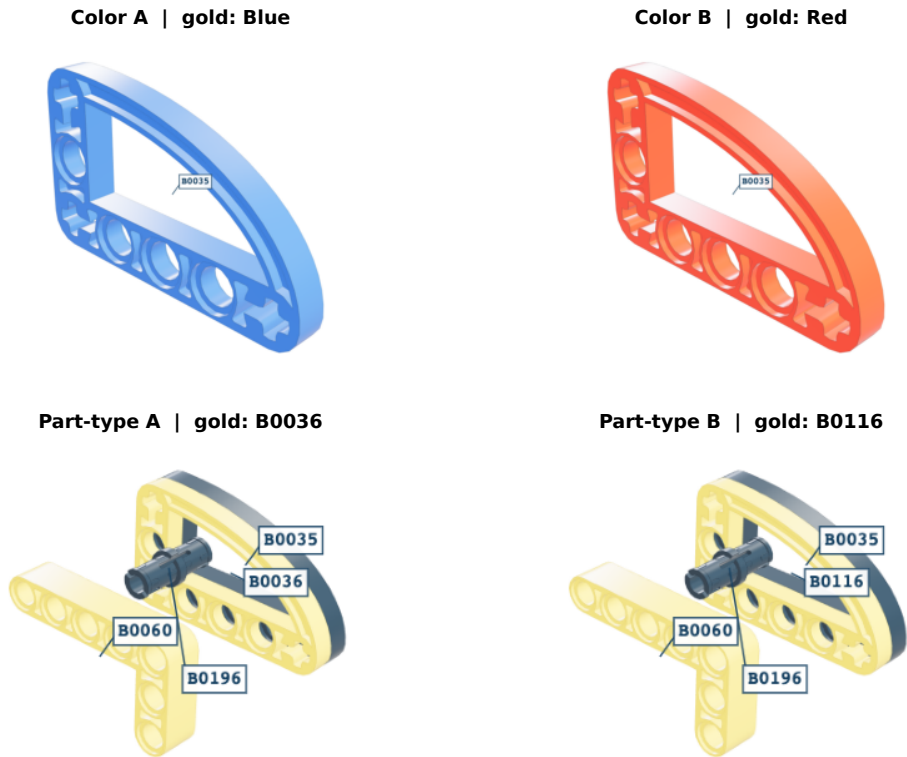


Figure 1. Actual paired observations from OMR 42004, shown with identical within-family paper crops. Top: Color asks for the current color of B0035; only the rendered body-color parameter changes, from Blue to Red. The original source color remains Black in provenance. Bottom: Part-type keeps target B0035 fixed and swaps candidate references B0036 and B0116. The matching geometry is unchanged; the answer changes from B0036 to B0116. Bottom details omit the remote swapped candidate. Gold annotations and crops belong only to this figure. Model inputs are the full 1280×800 PNGs and new display-grounded questions. These are stimulus examples, not model response cases or human answerability judgments.

2. Related Work and Measurement Scope

Assembly benchmarks. BrickNet [6] represents placed brick instances and connectors in graph-backed generative assembly programs. LEGO-Puzzles [11] evaluates spatial/sequential reasoning in editable scenes with three-annotator validation in its pinned v3 protocol. PhyBlock [8] studies physical planning with dependencies and simulation. Numeric IDs and reusable scene assets are established machinery; our endpoint concerns two elementary visual properties, not assembly planning.

Contrastive and consistent answers. Contrast sets probe local decision boundaries using small meaningful input changes [2]. VQA v2 pairs similar images that support different answers to the same question, reducing the utility of language-only shortcuts [3]. Cycle-consistency and VQA-Rephrasings address answer stability under linguistic variation [10]. Required change and required invariance differ: a constant answer can be stable but incorrect.

Direct paired-perturbation neighbor. VisualFLIP [14] minimally changes task-critical image evidence under a fixed question, with different golds. Its independent calls yield pair accuracy and competence-conditioned Collapse Rate; a separate sequential diagnostic exposes prior-answer persistence. Its v1 preprint reports 1,374 images, four perturbation categories and 24 MLLMs. It includes controlled rendering and manual pair audit. Our narrower LDraw study specifies source/display identity, finite joint assignment with declared oracle-rule bounds, and exact-wire review/receipt lineage. Those particular contracts are not specified in the inspected VisualFLIP construction/evaluation sections.

Reference grounding and construct validity. Numeric labels are narrower than referring-expression comprehension [9] or Winoground’s joint image–caption matching [12]. Small objects, resolution and off-construct requirements can explain grounding failures [1]; independent review therefore judges exact questions and pixels, including

109 visual answer uniqueness.

110 **Controlled visual and graph evidence.** CLEVR [5] and
111 GQA [4] use explicit scene structure and question opera-
112 tions. Our source structure supports provenance but stays
113 outside visual requests. The separate graph audit instanti-
114 ates a known limitation: six-cycle/two-triangles pairs are
115 indistinguishable to anonymous uniform-feature 1-WL [13].
116 This is not a new impossibility result or a trained-GNN ex-
117 periment.

118 3. Resource and Reference Contract

119 3.1. Preserved sources and separate roles

120 The resource preserves 617 questions, 15,334 placed LDraw
121 parts and 24 human-authored assemblies [7], including origi-
122 nal files, transforms and answers. The 140 visual parents
123 define display-v3 observations; 477 other historical items
124 remain separate controls and structural tasks. Parent IDs
125 link provenance, not transferable responses. Visual ques-
126 tions receive image/text; structural questions receive ex-
127 plicit graphs. These roles do not form a perception-to-action
128 chain.

129 Source bands D1–D4 describe assembly scale, not cog-
130 nitive difficulty. The supplement lists every source; the
131 separate dossier lists all families. The stable schema key
132 `shape-match` displays as Part-type.

133 3.2. Four meanings that must remain distinct

134 For a source hash h and instance ID i , canonical identity
135 $c = (h, i)$ identifies the original record. Source truth $T(c)$
136 stores its type, color and transform. In observation o , a dis-
137 play mapping $d_o(\ell)$ selects the visible referent of label ℓ .
138 The scored answer y_o is a function of the question and this
139 observation, not necessarily of $T(c)$ alone.

140 Every new visual question includes the following rule:
141 *References select objects in the supplied image only. A la-*
142 *bel such as B0001 is an image-local display reference, not*
143 *a request to retrieve a historical source instance. Renders*
144 *may have edited appearances or label assignments. Answer*
145 *only from the current image.* This rule is part of the hashed
146 model-visible text on both sides. The evaluator retains c ,
147 $T(c)$ and the display mapping outside the request.

148 Color asks, “What color is the object currently marked
149 B0035 in this image?” Part-type asks, “Which displayed
150 candidate label currently marks an object with the same vis-
151 ible part geometry as the target marked B0035? Ignore color
152 and pose.” The complete Part-type prompt also repeats that
153 the current image, rather than historical identities, deter-
154 mines the answer. Options have explicit choice IDs. The
155 response contract requires exactly one JSON object contain-
156 ing a listed `choiceId`.

3.3. What is physically preserved

Source geometry and poses are immutable. Isolated or
operand-only views may omit supports; they are observa-
tions, not freestanding configurations. Color edits and sepa-
rate part panels propose no new physical assembly.

The supplied connector graph covers recognized rela-
tions, not every contact or stability condition. The histor-
ical inventory still contains 111 other flagged tasks and 573
intersection candidates requiring human decisions. A can-
didate may be intended mating, an approximate mesh inter-
section or a source defect. This revision does not resolve
those physical judgments through visual benchmark code.

4. Interventions and Exact Observations

4.1. Color: appearance sanity track

The 73 Color pairs use the same original geometry, source
pose, camera, scale, label projections, visibility and mate-
rial settings. Only the body-color parameter changes be-
tween two distinct named opaque colors. Both arms ex-
plicitly permit edited appearance; neither arm asks for the
original source color. The 146 observations use Red, Blue,
Yellow and Black, with aggregate counts 37, 37, 36 and 36
respectively. These are displayed gold categories, not a sam-
ple of natural material colors.

All 146 images contain one rendered operand and one la-
bel. Color is an *appearance sanity track*: it tests following
current rendered color, not reference selection among can-
didates. A frozen whole-image color baseline, given only
one PNG and its text/options, correctly answers 141/146
endpoints, with 5 abstentions and 68/73 Both. It uses no
private mask or source record.

The machine audit checks common camera and label
metadata and compares the two native images. All 73 pairs
have zero changed pixels outside the common projected
body mask dilated by one antialiasing pixel. Independent
regeneration of three canary observations is byte-identical.
These checks locate the pixel change; they cannot estab-
lish that humans perceive every chosen category unambigu-
ously.

4.2. Part-type: candidate-reference reassignment

The 67 Part-type pairs retain the target geometry and token.
The matching candidate exchanges labels with one other
candidate, so the current reference that selects matching ge-
ometry changes. The option list remains fixed within the
pair. Source part numbers independently derive a candidate
gold, but the question asks for visible geometry. Similar
types, hidden features or a confusing label leader can de-
feat visual uniqueness even when the record-level match is
exact.

Both Part-type arms receive new option orders, image
renders, observation IDs and review lineage. Base geometry

Table 1. Choice-ID gold counts beside the planned experiment inventory. Color is frozen; only Part-type was optimized. Both families cover 24 sources. Actual human qualification is 0/280 primary endpoints.

Family	Arm	A	B	C	D
Color	A	13	22	19	19
Color	B	16	15	20	22
Part-type	A	16	17	17	17
Part-type	B	17	17	17	16

pixels match exactly within each pair; changed pixels lie within the label masks. Two regenerated canaries are byte-identical. In Figure 1, B0035 stays fixed while the matching candidate’s label changes from B0036 to B0116. No old Part-type review or response is transferable.

4.3. Joint assignment and bounded shortcut audit

For each parent we enumerate all four-option permutations and all nonmatching swap candidates already visible in the operand set, with an on-screen anchor and exactly one source-type match. The 4,824 configurations reduce to 2,412 constraint-equivalent representatives. Binary selection chooses one configuration per parent. Bounds are committed before optimization: each arm has 16 or 17 golds per position, each off-diagonal transition has 5 or 6 pairs, and diagonal counts are zero. Every fixed-choice predictor must score at most .30 source-macro on either arm. Thirteen declared oracle-A rules must have B accuracy at most .40 in both micro and source-macro estimates.

The objective minimizes summed integer SHA-256 configuration costs; sorted inputs and the checked single-thread HiGHS runtime give the recorded optimum 1,007,668. The immutable assignment is authoritative across runtimes. Rules use option order, alternate label numbers or nearest/farthest A-side source centers and projected anchors; none reads B pixels. Independent gold/feature reconstruction enforces the contract before sealing, live inference and publication. These are design bounds, not significance thresholds or evidence against every possible shortcut.

Tables 1 and 2 expose actual priors. The oracle-A first-alternative audit has 23/67 Both (34.33% micro; 26.94% source), conditional on supplied A gold. This is neither a model ceiling nor evidence of protocol-legal leakage. Because each transition cell is at most 6, any single deterministic mapping from A choice alone to B choice has in-sample accuracy at most $4(6)/67 = 35.82\%$. This restricted ceiling does not apply to predictors using labels, source IDs or additional features. Color was not reoptimized: its last-alternative and cyclic previous rules reach 32/73. Balanced semantic color totals therefore do not imply balanced choice transitions or a shortcut-free family.

A separate set of 67 position-reference observations

Table 2. Deterministic B accuracy (%), micro and equal-source. [†]Oracle supplies correct A; these rules read no B image, so their B accuracy equals Both. Fixed choices have Both=0. These are stimulus audits, not learned-model scores. All 13 Part-type oracle rules, intervals and individual errors are released.

B predictor	Color		Part-type	
	Micro	Source	Micro	Source
Fixed A	21.92	18.47	25.37	20.28
Fixed B	20.55	20.76	25.37	27.71
Fixed C	27.40	31.39	25.37	27.50
Fixed D	30.14	29.38	23.88	24.51
First alternative [†]	26.03	23.47	34.33	26.94
Last alternative [†]	43.84	43.12	32.84	32.01
Cyclic next [†]	30.14	31.46	35.82	38.68
Cyclic previous [†]	43.84	43.61	34.33	28.89
Cyclic two [†]	26.03	24.93	29.85	32.43

shows a target and four candidate panels without text inside the image. Options refer to candidate positions. This provides a geometry-matching comparison under a different reference interface. Panel layout and visual access also change. This control cannot isolate geometry vs. position; it is descriptive only. It has separate human reviews and never enters primary paired Both.

4.4. Freezing the input boundary

An exact observation includes the complete system/user messages, question, options, native 1280×800 PNG bytes, resolution and image-detail policy. All 347 observations are frozen before review: 280 primary endpoints and 67 auxiliary controls. Rendering metadata, canonical source truth, conditions, pairing and gold are evaluator-only records. Neither endpoint is supplied as conversational history to the other.

Each run separately pins the model revision, adapter and generation settings and hashes its HTTP request. That envelope must reproduce the reviewed messages and PNG bytes; internal provider preprocessing remains unobserved. Changes to text, pixels, dimensions or image policy require new review. Envelope-only changes do not create a new visual question.

5. Human Qualification and Response Analysis

5.1. Independent review of both sides

Six distinct reviewers receive 115, 116, 114, 115, 115 and 119 units, respectively, in shuffled queues with opaque unit IDs. Every observation receives two judgments, but nobody sees more than one observation of a parent, including the auxiliary panel. Thus A/B reviewer sets have empty intersection; reviewer slots do not identify an arm. The interface shows one observation at a time, its exact text and scroll-

lable native pixels. It withholds the condition name, parent/pair membership, source record, gold, model output and the other reviewer’s decision. The primary denominator is always 280; it cannot become a B-only queue through task-ID deduplication.

A raw review contains the reviewer pseudonym, timestamp, categorical decision, visual choice, reason, and attestations of native-size inspection and independence. Four structured checks ask whether the target reference is unique, exactly one option is visually correct, visibility is sufficient, and the observation lacks reference contradictions. Decisions distinguish decidable, ambiguous, unanswerable and contradictory cases.

Matching decisions, choices and checks produce a final judgment. Any disagreement among these fields requires explicit adjudication. Up to three additional people occupy adjudicator slots; none is an initial reviewer or sees multiple observations of a parent. Assignment, imports and final reconstruction all verify this exposure rule. Both original records and raw adjudication exports remain intact. A final decidable choice that conflicts with the generated gold blocks finalization and requires revising the item and repeating QA, rather than silently replacing the answer. Agreement reports include raw decision agreement, multiclass Cohen’s kappa, choice/check agreement and their actual denominators. Undefined kappa remains undefined.

The runner requires final decisions for every planned endpoint before real visual inference. Nondecidable final judgments remain in the full denominator; only pairs with two final decidable endpoints enter the additional QA subset. This prevents post-result item selection. The gate recomputes decisions from retained raw files. It verifies recorded provenance and attestations, not a person’s identity through cryptography. Coordinator oversight remains necessary.

5.2. Correct adaptation and failure denominators

For pair i , let p_{Ai}, p_{Bi} be normalized choices, $y_{Ai} \neq y_{Bi}$ their golds, and v_{Ai}, v_{Bi} indicate valid output. Invalid or missing predictions are never equal to a gold. For N fixed pairs, we report

$$AAcc = N^{-1} \sum_i [p_{Ai} = y_{Ai}], \quad (1)$$

$$NewAcc = N^{-1} \sum_i [p_{Bi} = y_{Bi}], \quad (2)$$

$$Both = N^{-1} \sum_i [p_{Ai} = y_{Ai}][p_{Bi} = y_{Bi}]. \quad (3)$$

The B outcomes partition into new-gold correctness, old-gold retention, another valid choice and invalid output. In particular,

$$R_{old} = N^{-1} \sum_i v_{Bi} [p_{Bi} = y_{Ai}]. \quad (4)$$

Old-gold retention is a behavioral outcome, not proof of memory or an internal causal mechanism.

Conditional adaptation is Both divided by A accuracy for the micro estimate. Its denominator is the number of A-correct pairs; a zero denominator yields null. Conditional source-macro analysis averages within-source conditional rates only over sources with eligible pairs and reports that coverage. We retain unconditional estimates so conditioning cannot hide a weak A endpoint.

Format-valid rates are reported for A, B and both endpoints, alongside semantic rates. Timeouts, API failures, refusals, missing receipts and illegal formats remain incorrect in the planned denominator, with separate failure reasons. A stable invalid string is not semantic consistency. The auxiliary position control reports its own accuracy and validity without being pooled with either visual family.

5.3. Aggregation, selection and comparisons

For a pair outcome z_{si} from source s , the primary estimator is

$$\hat{\mu} = \frac{1}{|S|} \sum_{s \in S} \frac{1}{n_s} \sum_{i=1}^{n_s} z_{si}. \quad (5)$$

Each family has one primary pair per parent. We also report micro rates, pair counts and source counts. A family absent from a source does not contribute an artificial zero. Percentile intervals use 10,000 whole-source bootstrap draws with seed 20260920. Resampling sources preserves within-source dependence. We also freeze 18 primary author groups and 17 contributor-connected groups from the source credits. Sensitivity reports include equal-author means, author-cluster intervals for both weighting schemes, and leave-one-author-out differences. Contributor clustering also includes OMR editors. All 276 source-pair part-inventory Jaccards are published; inventory overlap is not aligned geometry. A direction changing under author omission limits any cross-source stability claim.

Full and both-decidable estimates are published together with exclusion reasons and source coverage. The QA-minus-full difference resamples the same source weights for both estimates, recomputing each mean on its eligible sources. This retains changes in source composition. Overlapping exclusion reasons are not added as independent counts. Empty subsets and bootstrap draws without selected sources are explicitly recorded.

Model comparisons use paired outcomes on the same items. Ordinary-A, NewAcc and Both ranks use exact rational point comparisons and descending midranks for ties. A reversal means the signs of the A and Both differences disagree; a tie is separately classified. Paired source intervals accompany the differences. We also record whether comparison directions agree between full and QA subsets. The intervals are descriptive and unadjusted for multiple comparisons. An interval containing zero does not support a

Table 3. Frozen complete roster. OpenAI is the provider for every row. Runs means completed/planned; each plan includes visual, no-image and text-graph conditions. No collection dates or scores exist yet. The complete inclusion and failure policies are frozen in the roster.

Model	Pinned revision	Runs
GPT-4.1	gpt-4.1-2025-04-14	0/3
GPT-4o	gpt-4o-2024-08-06	0/3
GPT-4o mini	gpt-4o-mini-2024-07-18	0/3

directional ranking. The estimand is one frozen invocation set, not cross-call stability; temperature zero does not establish provider determinism.

5.4. Frozen roster and image-dependency control

Table 3 specifies three OpenAI snapshots, chosen to compare an instruction-following reference, an earlier multi-modal model and its smaller tier under one implemented transport. This is not a current-model or cross-provider leaderboard. Each has one 347-observation visual run, one 347-observation no-image control and one 219-observation graph run: exactly nine runs and 2,739 scheduled calls. The no-image condition removes only image content, retaining question/options. All 140 pairs then have byte-identical requests and different golds. Identical deterministic responses force Both=0 and symmetric endpoint accuracy at most 1/2. This is an image-content floor, not a reference-selection floor.

All adapters pin Chat Completions, exact requested/returned revisions, high-detail native images, temperature zero, 128 output tokens, a 90-second timeout and one attempt with no retries. Official model documentation was checked; live account access remains untested. Analysis requires exact equality of observed and planned run sets. Every entry binds adapters, requests, raw receipts, QA and report rows through a model-run index. Missing or extra runs block empirical tables. Interrupted runs require explicit closure, retaining pre-call intents and missing-attempt records in the original denominator. They cannot be replaced by a favorable repeat.

5.5. What would establish diagnostic value?

A useful result would reveal differences obscured by ordinary accuracy, robust to answer priors, image withdrawal, QA selection and author dependence. If Both merely reproduces A rankings or differences reflect format/API failures, added diagnostic value is limited. Deterministic case extraction links the first hash-ordered QA-decidable example of each response transition to its requests and raw receipts; absent types remain absent. No learned rankings, reversals or response cases exist yet. Table 4 records evidence availability.

Table 4. Current primary visual inventory. QA final counts completed human decisions, not rendered images. Completed model runs are 0/9 planned. Both-decidable coverage is 0/140 pairs; this is an evidence-availability count, not model accuracy.

Family	Pairs	Sources	QA final
Color	73	24	0/146
Part-type	67	24	0/134

6. Graph Audit: Which Controls Resist Shallow Rules?

6.1. Natural evidence and hypothetical partitions

A historical Graph deletion question asks for the number of connected components after deleting a specified node from an explicit undirected graph, including isolated survivors. Duplicate edges are deduplicated. Independent BFS and union-find implementations agree on all golds. These algorithms validate the supplied graph contract, not the completeness of the physical connector detector.

The audit retains all 73 natural questions from 24 sources. It also reclassifies the existing 219 hypothetical observations: each parent has an anchor, an answer-changing partner and an isomorphic answer-preserving partner. The 12 strong change pairs span eight sources and match surviving node set, edge count, target degree, complete surviving degree multiset and isolate count. The other 61 change pairs span 23 sources and are explicitly called *degree-visible controls*. Their invariance pairs are reported separately. The shared anchors mean these four partitions cannot be summed as independent observations.

Strong pairs compare a six-cycle with two triangles, with identical extra isolated nodes. Weak pairs use a path versus a triangle plus isolates. Both templates can match node and edge counts, but only the former matches degrees. These are hypothetical edge lists derived for a controlled test; they are not claims about newly discovered physical connections in the source assemblies.

6.2. Baselines and matching limits

We include node-count, edge-count, degree-histogram and anonymous 1-WL feature lookups; the isolates-plus-one rule; and BFS/union-find. For a surviving graph G' , the heuristic is

$$\hat{c}_{\text{iso}} = n_0 + \mathbf{1}[|V(G')| > n_0], \quad (6)$$

where n_0 is its isolate count. It does not traverse components. Feature lookups use labels only from other sources, select the most frequent answer, break ties toward the smaller integer, and abstain on unseen features. Abstentions count as incorrect. Training uses changed endpoints once;

Table 5. Computed algorithm outputs, not learned-model results. Natural correct items and hypothetical both-correct pairs use distinct source-held-out fits. Strong change is a small proof of concept; degree-visible controls support a weaker claim. Invariance reuses anchors.

Algorithm	Natural correct / 73	Strong change Both / 12	Degree-visible Both / 61	Strong invariance Both / 12
Node-count lookup	63	0	0	6
Edge-count lookup	44	0	0	7
Degree-histogram lookup	69	0	61	6
Anonymous 1-WL lookup	69	0	61	6
Isolates + one	72	0	61	6
BFS	73	12	61	12
Union-find	73	12	61	12

invariance copies do not duplicate training labels. Natural and hypothetical fits remain separate.

Anonymous uniform-feature 1-WL gives identical signatures on all strong change pairs. A deterministic predictor restricted to that signature must give the same answer to different golds, hence Both=0. This also limits message passing no more expressive than 1-WL [13]. Unique IDs, positions, spectral and higher-order methods are outside the claim; no trained GNN result is implied.

6.3. Measured results and uncertainty

Table 5 shows that isolates-plus-one obtains 72/73 natural correct answers, compared with 73/73 for BFS. Its natural source-macro accuracy is 97.92%, with a descriptive 95% source interval [93.75, 100]. Degree-histogram and 1-WL lookups reach 69/73. Natural accuracy barely distinguishes traversal from these shallow rules.

On degree-visible change pairs, degree-histogram lookup, 1-WL lookup and isolates-plus-one all achieve 61/61 Both. On strong change pairs, each obtains 0/12; BFS and union-find obtain 12/12. Strong invariance illustrates another distinction: the degree and 1-WL predictors are consistent across the two isomorphic observations, but only 6/12 pairs are jointly correct. Invariance of predictions alone would reward the six stable wrong pairs.

Strong change directions have six increases and six decreases, although raw arm histograms differ. Unordered Both and symmetric endpoint accuracy retain the 12-pair denominator. Histograms, all strong pairs and invariance results appear in the supplement.

Approximate pair Wilson intervals are [0, .2425] for 0/12 and [.7575, 1] for 12/12; pair independence is imperfect. Whole-source intervals can degenerate at boundary outcomes. Neither establishes generality beyond eight sources and one template.

7. Reproducibility, Limits and Empirical Acceptance

An earlier oracle-assisted rule supplied A gold obtained 67/67 Both. This motivated balancing; it was not a legal single-observation attack. Archived release bytes and render assets are hashed. New Part-type IDs prohibit transferring old reviews/responses. Analysis reparses receipts and checks served revisions, roster, gold and common QA lineage. Dry runs and fixtures cannot enter empirical reporting.

Machine verification covers 347 visual observations, gold/shortcut reconstruction, pixel audits, 2,739 planned envelopes and 27 v3 protocol tests, with 26 historical tests preserved. The draft adds 11 audit tests and complete 566-observation ledgers. These establish implementation behavior; human agreement and provider authenticity require genuine records and collection oversight.

The release has zero genuine human reviews and 0/9 completed model runs. The primary empirical question remains unresolved. Actual judgments, full/QA results, retention, format effects and genuine cases are required.

The supplement details protocols, sources and strong pairs. The 617-item dossier links to 3D rotation, zoom, expansion and action replay; inspection views do not substitute for native-input human qualification.

The bounded scope is instance-label-grounded evidence following on two atomic visual attributes and 24 sources. Rendered colors do not cover natural color constancy; candidate-label changes do not establish general referring-expression understanding. The auxiliary panel comparison changes visual access. Graph evidence concerns a limited recognized graph and a small hypothetical challenge. Public source reuse also leaves contamination and cross-source dependence concerns. These limits bound what future positive results could establish.

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